October University for Modern Sciences and Arts (MSA)
Department of Electronics and Communication
Course: ECE552 Mobile Communication Systems
Instructors: Prof. Fawzy Ibrahim Dr. Abdelrahman El-Akhdar
Assoc. Prof. Nariman Abdelsalam

Problem Set #1 Solution

Principles of Wireless Communications

Question #1.1

Given the frequency band assigned to a cellular phone network from 880 to 915 MHz for uplink, and from 925 to 960 MHz for downlink, If the user channel bandwidth, $BW_{ch} = 200$ kHz, each channel can be divided in 8 time slot one for each user and the network contains total number of clusters, $N_{cls} = 3000$. Find

a) Total # of channels in the mobile network system and cluster (N_{chnet} & N_{chcls}).

b) Number of channels in each cell, N_{chcell}, for cluster sizes, N = 4.

c) Number of users in each: cell (N_{uscell}), cluster (N_{uscls}) and the mobile network (N_{usnet}).

Solution:

a) The total number of channels in the network, $N_{chnet} = BW / BW_{ch}$.

so $N_{chnet} = 35 \times 10^6 / 200 \times 10^3 = 175$ channel.

The Number of channels per cluster, N_{chcls} = total number of channels in the network, $N_{chnet} = 175$ channel.

b) Number of channels per cell = $N_{chcell} = N_{chcls} / N = 175 / 4 = 43.35$ channel.

c) Number of simultaneous users per cell (Cell Capacity), $N_{uscell} = 43 \times 8 = 344$ user.

Number of simultaneous users per cluster (Cluster Capacity), $N_{uscls} = N_{uscell} \times N = 344 \times 4 = 1376$ simultaneous user.

The mobile network capacity, $N_{usnet} = N_{uscls} \times N_{cls} = 1376 \times 3000 = 4.128$ million simultaneous user.

Question #1.2 **[HW]**

Given the frequency band assigned to a cellular phone network from 1850 to 1910 MHz for uplink, and from 1930 to 1990 MHz for downlink, If the user channel bandwidth, $BW_{ch} = 200$ kHz, each channel can be divided in 8 time slot one for each user and the network contains total number of clusters, $N_{cls} = 2700$. Find

a) Total # of channels in the mobile network system and cluster (N_{chnet} & N_{chcls}).

b) Number of channels in each cell, N_{chcell}, for cluster sizes, N = 7.

c) Number of users in each: cell (N_{uscell}), cluster (N_{uscls}) and the mobile network (N_{usnet}).

Solution:

a) The total number of channels in the network, $N_{chnet} = BW / BW_{ch}$.

so $N_{chnet} = 60 \times 10^6 / 200 \times 10^3 = 300$ channel.

The Number of channels per cluster, N_{chcls} = total number of channels in the network, $N_{chnet} = 300$ channel.

b) Number of channels per cell = $N_{chcell} = N_{chcls} / N = 300 / 7 = 42$ channel.

c) Number of simultaneous users per cell (Cell Capacity), $N_{uscell} = 42 \times 8 = 336$ user.

Number of simultaneous users per cluster (Cluster Capacity), $N_{uscls} = N_{uscell} \times N =$ $336 \times 7 = 2352$ simultaneous user.

The mobile network capacity, $N_{usnet} = N_{uscls} \times N_{cls} = 2352 \times 2700 = 6.350$ million user.

Question #1.3

A mobile service area of 50 x 30 km need to be covered, the cluster size N = 12, cell radius is R = 5 km. Compute the network capacity in the covered area in terms of number of simultaneous mobile users given that:

a) Uplink band: 1850 – 1910 MHZ, Downlink band: 1930 – 1990MHz.

b) Each half duplex channel user is 200 KHZ BW and (8 TDM slots are applied).

Solution

Service area, A =50 x 30 = 1500 km^2

Cell area $A_{cell} = 2.6 \times R^2 = 2.6 \times (5)^2 = 65\ km^2$

For N=12

Cluster area = 12 x Cell area = 12 x 65 =780 km^2

clusters in the area = N_{cls} =1500 / 780 = 2 clusters.

Total number of channels, N_{chnet} = 60 / 0.2 = 300 channels.

#users per cell, N_{uscell} = (300 / 12) x (8) = 200 user.

#users per cluster, N_{uscls} = 200 x 12 = 2400 user.

Mobile system capacity in the service area, N_{usnet} = 2400 x 2 = 4,800 users.

Question #1.4 **[HW]**

A mobile service area of 25 x 15 km need to be covered, the cluster size N = 7, cell radius is R = 2 km. Compute the number of simultaneous users in this network given that the total number of RF channels, N_{chnet} = 1000 channel and the cell area for hexagonal, $A_{cell} = 2.5981 \times R^2$.

Solution as done in Question #1.3.

Question #1.5

For isotropic antennas working at frequency, f = 1.8 GHz, if the received power at a reference distance, d_o = 0.5 km, is equal to $P_r(d_o)$ = 1 μW, find the received powers at distances, d = 1 km, 2,5 km, 5 km and 10 km from the same transmitter using Mobile Environment Model for the path loss exponent or environment factor, n = 3.

Solution

For Mobile Environment Model, the received power at any point at a distance $d > d_0$ can be obtained using the following equation:

$$Pr\,(d)dBm = 10\log[\,Pr\,(d0\,)] - 10n\log(d\,/\,d0\,)$$

P_r (1 km) = 10log (10^{-6}) – 30 log(2) = - 69.03 dBw. P_r (2.5 km) = 10log (10^{-6}) – 30log (5) = - 80.9 dBw. P_r (5 km) = 10log (10^{-6}) – 30log (10) = - 90 dBw.

P_r (10 km) = 10log (10^{-6}) – 30log (20) = - 99.03 dBw

Question #1.6 **[HW]**

For isotropic antennas working at frequency, f = 1.8 GHz, if the received power at a reference distance, d_o = 0.5 km, is equal to $P_r(d_o)$ = 1 μW, find the received powers at distances, d = 1 km, 2,5 km, 5 km and 10 km from the same transmitter using Mobile Environment Model for the path loss exponent or environment factor, n = 4.

Question #1.7

In a cellular mobile system, if cell A and cell B are co-channel cells as shown in Fig. 2.1, the minimum signal-to-interference ratio required to achieve appropriate transmission is S/I =15 dB, and if path loss exponent is n = 2.5. Find,

a) The frequency reuse factor, N.
b) The signal-to-interference ratio (S/I) in dB for a mobile receiver located in cell A considering only first tier of interfering cells, $n_o = 6$.
c) Is the calculated frequency reuse factor sufficient to achieve the required S/I? If not, what is the suitable value for N to achieve required S/I?
d) If 20 MHz of total spectrum is allocated to provide a duplex wireless cellular system and each simplex channel has 25 kHz RF bandwidth, find the number of duplex channels, the total number of channels per cell site and the cluster capacity.

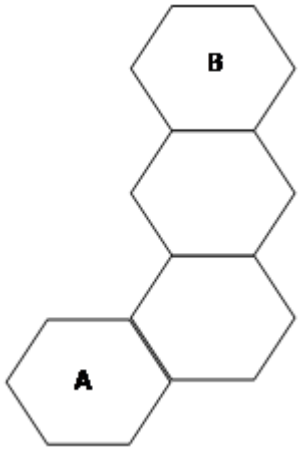


Fig. 2.1

Solution

S/I =15 dB

a) the cluster size $N = \left(i^2 + j^2 + ij\right)$

From Fig.1, $i = 1 \; and \; j = 2$

$$N = (1)^2 + (2)^2 + 1 \times 2 = 7$$

b) $S/I = (D/R)^n / n_0 = \left(\sqrt{3N}\right)^n / n_0$

n = 2.5

Since first tier of interfering cells is only considered, n_0=6

$$\frac{S}{I} = \frac{\left(\sqrt{3N}\right)^n}{n_0} = \frac{\left(\sqrt{3 \times 7}\right)^{2.5}}{6} = 7.492$$

$$\left(\frac{S}{I}\right)_{dB} = 10 \log 7.492 = 8.746 \; dB$$

c) The calculated frequency reuse factor $N = 7$ is not sufficient to achieve the required S/I as $\left(\frac{S}{I}\right)_{acheived} = 8.764 \; dB < \left(\frac{S}{I}\right)_{required} = 15 \; dB$.

$$\left(\frac{S}{I}\right)_{dB} = 10\log\frac{S}{I} = 15$$

$$\frac{S}{I} = 31.623$$

$$31.623 = \frac{\left(\sqrt{3N}\right)^{2.5}}{6}$$

$$N = 22.151$$

So N should be at least 22.151
We select $N = 27, (i = 3, j = 3)$

d) System bandwidth =BW = 20MHz.
Channel bandwidth, BW_{ch}= 25 kHz × 2 simplex channels = 50 kHz/duplex channel.
Total number of duplex channels in the network, N_{chnet}= BW / BW_{chFDD}.
soN_{chnet} = 20 x 10^6 / 50 x 10^3 = 400 Channels

Total number of channels per cell site= N_{chcell}= N_{chnet}/ N = 400 / 27 = 14.814 channels (14simultaneous users).

Cluster capacity = Number of channels per cluster, N_{chcls} = Total number of duplex channels in the network= 400.

Question #1.8 **[HW]**

In a cellular mobile system, if cell A and cell B are co-channel cells as shown in Fig. 2.2, the minimum signal-to-interference ratio required to achieve appropriate transmission is S/I =18 dB, and if path loss exponent is n = 3. Find,

a) The frequency reuse factor.
b) The signal-to-interference ratio (S/I) in dB for a mobile receiver located in cell A considering only first tier of interfering cells.
c) Is the calculated frequency reuse factor sufficient to achieve the required S/I? If not, what is the suitable value for N to achieve required S/I?
d) If 25 MHz of total spectrum is allocated to provide a duplex wireless cellular system and each simplex channel has 200 kHz RF bandwidth, find the number of duplex channels, the total number of channels per cell site and the cluster capacity.

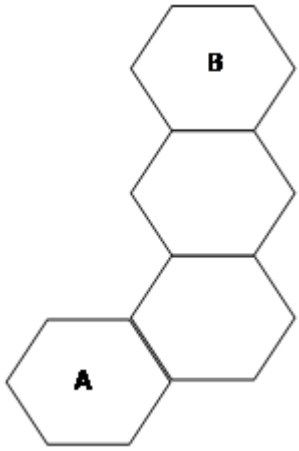


Fig. 2.2

Question #1.9

In a GSM network, cell A_1 is the desired cell and A_2 is a co-channel cell as shown in Fig. 2.2 with cluster size, N = 12. For a mobile located in cell A_1, find the minimum cell radius R to give a forward link signal-to-interference ratio, S/I ≥ 30 dB. Assume the following:

- Co-channel interference is due to base A_2 only.
- The carrier frequency, f_c = 900 MHz.
- Reference distance, d_o = 1 km (assume free space propagation from the transmitter to d_o).
- Assume Omni directional antennas for both transmitter and receiver, where G_{base} = 10 dBi and G_{mobile} = 3 dBi.
- Transmitter power, P_t = 20 W (assume equal power for all base stations).
- Assume log-normal path loss model is applied between the mobile and base A_1, with loss exponent, n_1 = 2 and A_2 with loss exponent, n_2 = 3.5.

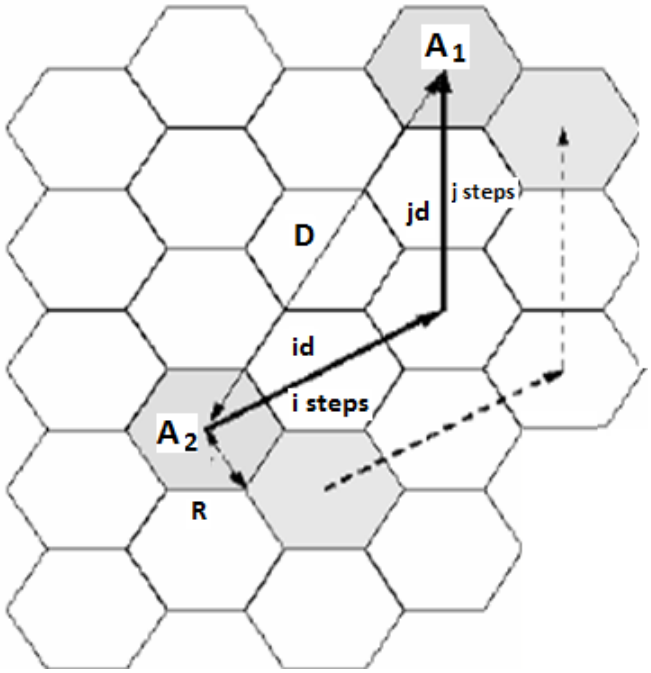


Fig. 2.3

Solution

Given P_t = 20 W , $G_{base} = G_t$ = 10 dBi = 10, $G_{mobile} = G_r$ = 3 dBi = 2 and d_o = 1 km.

$$\lambda = c / f_c = 3x10^8 /(900x10^6) = 0.33 \ \ m$$

For free space propagation the received power at d_o, $P_r(d_o)$ is given by Eq.(2.5):

$$P_r(d_0) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d_0^2} = \frac{20x10x2x(0.33)^2}{(4\pi)^2 (1000)^2} = 2.76x10^{-7} \ W$$

$$P_r(d_0) = 10 \ \log \ (2.76x10^{-8}) = -65.6 \ \ dBm$$

- At the boundary of the cell A_1, the received power due to the transmitter A_1 is:

$$P_r(R)dBm = P_r(d_0) - 10n_1 \log(R/d_0) = -65.6 - 20\log(R/d_0)$$

- The distance between the cell A_1 and Cell A_2, is given by Eq. (2.5) as:

$$D = \sqrt{3}R\sqrt{i^2 + j^2 + ij} = \sqrt{3N}R = \sqrt{3x12}R = 6R$$

- At the boundary of the cell A_1, the received power due to the transmitter A_2 is:

$$P_r(R)dBm = P_r(d_0) - 10n_2 \log(d_1/d_0) = -65.6 - 35\log(5R/d_0) - 24.46$$

where

$$d_1 = D - R = 6R - R = 5R$$

- The signal-to-interference ratio, S/I is given by:

$$S/I = P_r(R) - P_r(d_1) = -65.6 - 20\log(R/d_0) - [-65.6 - 35\log(R/d_0) - 24.46]$$
$$= 24.46 + 15\log(R/d_0)$$

  - C/I =30 dB= 24.46 + 15log (R/d) so 15log (R/d)= 5.54
  - So, the cell radius, R =2.31 km.

Question #1.10 **[HW]**

In a GSM network, cell A_1 is the desired cell and A_2 is a co-channel cell as shown in Fig. 2.3. For a mobile located in cell A_1, find the minimum cell radius R to give a forward link signal-to-interference ratio, S/I ≥ 20 dB. Assume the following:

- Co-channel interference is due to base A_2 only.
- The carrier frequency, f_c = 9900 MHz.
- Reference distance, d_o = 1 km (assume free space propagation from the transmitter to d_o).
- Assume Omni directional antennas for both transmitter and receiver, where G_{base} = 30 dBi and G_{mobile} = 10 dBi.
- Transmitter power, P_t = 50 W (assume equal power for all base stations).
- Assume log-normal path loss model is applied between the mobile and base A_1, with loss exponent, n_1 = 2 and A_2 with loss exponent, n_2 = 2.5.

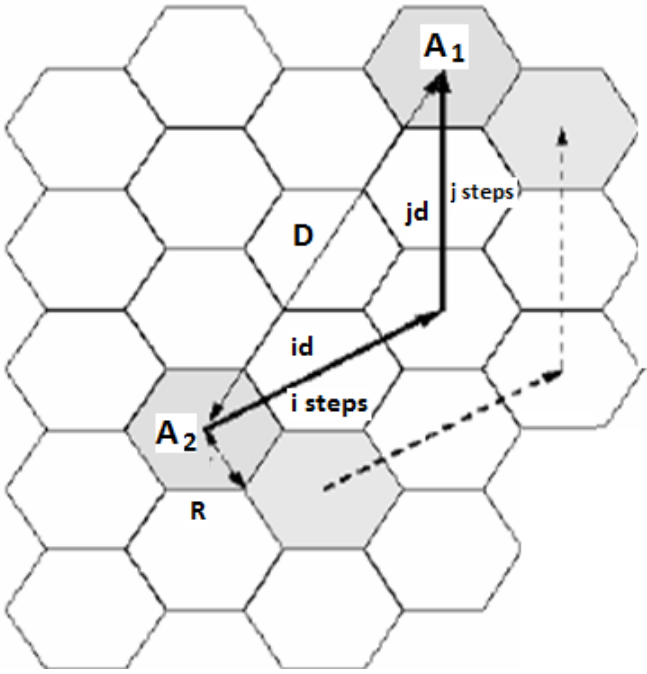


Fig. 2.3